

“Its finally here!!”—The First Generation (1G)

The first generation technology in mobile telephony began to arrive in the early 1980s. The primary tools used were the concept of cellular networks and analogue transmission using Frequency Division Multiple Access (FDMA) to separate calls from different users.

The FDMA technique assigns different frequencies for different calls to avoid conversations interfering with each other. Hence the first two terms—‘Frequency Division’, ‘Multiple Access’ meant that multiple users could use the same frequency at different times. Think FM Radio—each radio station broadcasts at its own frequency; similarly in FDMA, each call is on its own frequency.

The first use of First Generation technology was in the *Nordic Mobile Telephone* (NMT) system. While the technology was developed in Scandinavia by the Finnish Nokia (then Mobira) and Swedish Ericsson, it first rolled out in 1981 in, oddly enough, Saudi Arabia. The US and the rest of Europe would soon follow, with the *Advanced Mobile Phone Service* (AMPS) and the *Total Access Communication System* (TACS) respectively.

The systems proved to be quite robust; the NMT had excellent coverage of the unique terrain of the Scandinavian countries. In fact, the NMT and AMPS systems are still existent as backup networks, though they have been replaced by newer digital technologies. NMT even had a crude system for text messaging, called DMS—Data and Messaging Service.

Because the cellular operators were limited to a particular range of frequencies, there were only so many frequencies that could be allotted to calls before the entire frequency band was full. The AMPS network, for example, had a maximum capacity of 416 calls per cell—this when their system was gaining popularity. What if the 417th caller had an emergency?